Meaningful data makes a meaningful picture.

TEMPLATE

CLASS SITE

Wednesday 12:00–14:50 · E401 → PC Lab · Prof. DY Kwon · Use this for Week 7 practice and the Week 8 midterm

This week is about data: what you collect, and what the AI can make of it. You build a survey and an automation that turns all the answers into one report and one infographic — with real numbers like "4 of 5 customers". Today you practise with your own test answers.

	Week 7 · today — DATA	Week 8 · midterm — PEOPLE
Idea	Meaningful data → a meaningful picture	Connection with people: reach them, ask honestly
Ask yourself	What must I collect so the answer means something?	Who should I ask, and how, so they answer honestly?
You make	The survey + the automation, tested with your own answers	Real answers from 5+ classmates — and the value you find in them

What you will build — five points

Google Sheets Watch New Rows reads the answers → **Text aggregator** joins all answers into one text → **Make AI Toolkit** counts, then writes the report → **HTTP** Download a file draws the picture → **Gmail** sends report + picture to you

THE MOST IMPORTANT IDEA THIS WEEK

The AI and the picture only see **what goes into the Text aggregator**. Vague questions give vague answers. Vague answers give the AI nothing to count, and then the picture invents numbers. **A good survey is the real work.** The automation only carries it.

Today, step by step

1 Google Form (p.2) → **2** template + **your own** Google account (p.3) → **3** Sheets · Limit (p.4) → **4** Text aggregator (p.4) → **5** AI prompt (p.5) → **6** fun picture prompt (p.6) → **7** Gmail (p.7) → **8** test with 2 answers, Run once (p.7) · problems (p.8)

Template link: `eu1.make.com/public/shared-scenario/5FQXKKzrnDY` — scan the TEMPLATE code above, or open it from Week 7 on the class site. Then press **Create new scenario from template**.

1 Make a Google Form for your business

Go to forms.google.com → Blank form. Give it a clear title, for example *“Quiet Sandbag — Feedback Survey”*. Then write two parts.

Part 1 — About the business (the AI counts these)

Three rules for every question:

1. **One thing per question.** “How is the price and quality?” is two questions. Split it.
2. **Make it countable.** Use **Multiple choice** or **Checkboxes** for most questions. Then the AI can say “4 of 5”.
3. **Only ask what changes your decision.** If the answer would not change anything you do, delete the question.

Part 2 — For the picture (the image uses these)

- How old are you? — *18–25 / 26–40 / 40+*
- How would you like to appear in a picture? — *man / woman / prefer not to say*
- Where would you use this? — *home / office / gym / outdoors*
- What mood fits you? — *calm / energetic / funny / serious*

> **DO NOT ASK**
> Names, phone numbers, student IDs, photos, or anything that tells who a person is. The answers travel to an AI service.



Example form. Every question is multiple choice or checkboxes, so every answer can be counted.

5 to 7 questions is enough. A survey that takes one minute gets answered.

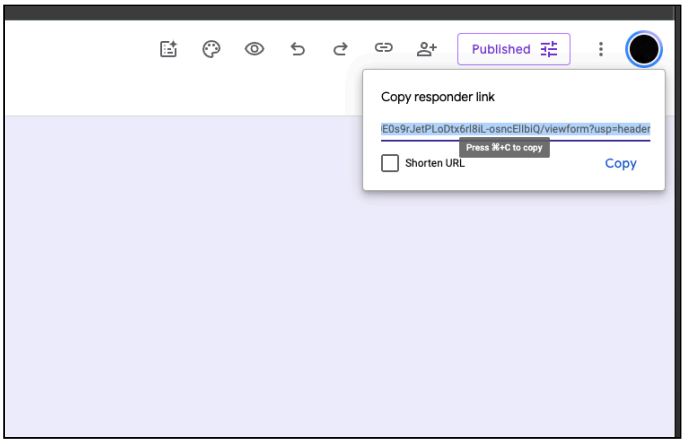
Connect the form to a Google Sheet

In the form, open the **Responses** tab → **Link to Sheets** → **Create a new spreadsheet**. Your answers now arrive in a sheet tab called **Form Responses 1**. Make will read this sheet.

Get the link you will send to classmates

Press **Publish** (top right). Then press the **link icon** → **Copy responder link**. This is the link your classmates open.

Open the link once yourself and look at it as a classmate will see it.



2 Copy the template — then make it yours

Open the template link (page 1) → press Create new scenario from template. Make copies all five points into your account.

The template has no account inside it. Every Google point needs YOUR account.

The template came from the professor's Make. His Google login does not come with it — and it must not. Until you connect your own account, the Sheets point and the Gmail point cannot work.

Connect your account — do this for Google Sheets and for Gmail

1. Click the point (Google Sheets, then Gmail).
2. At **Connection**, press **Add**.
3. A Google window opens. Sign in with **the same Google account** that owns your form and sheet.
4. Google shows what Make wants to do. **Tick every box** and press **Allow** / **Continue**. If you leave a box empty, Make will say "insufficient permissions" later.
5. Back in Make, your account name now appears in Connection. Press **Save**.

WHY THE PERMISSION SCREEN MATTERS

Make is asking Google: "May I read this person's sheets and send email as this person?" Only you can say yes. If the window does not open, allow pop-ups for make.com. On a phone this often fails — use the PC.

What you do NOT connect

- **Make AI Toolkit** — keep **Make's AI Provider (default)**. No key needed.
- **HTTP** — keep **No authentication**. Our class server does not need a key. It only needs your student ID (page 6).

Before you continue — check these four

Point	You should see
Google Sheets	Connection = **your** Google account
Make AI Toolkit	Connection = Make's AI Provider (default)
HTTP	Authentication type = No authentication
Gmail	Connection = **your** Google account (an orange "reauthorize until …" note is normal — ignore it)

WHY THIS IS AN "API" LESSON

Sheets and Gmail have apps inside Make, so you connect with a login. Our image server has no app — Make reaches it with **one web address** through the HTTP point. That is what an API is: a door you can call with an address. Some doors need a key (paid AI services). Ours needs only your ID.

3 Google Sheets · Watch New Rows

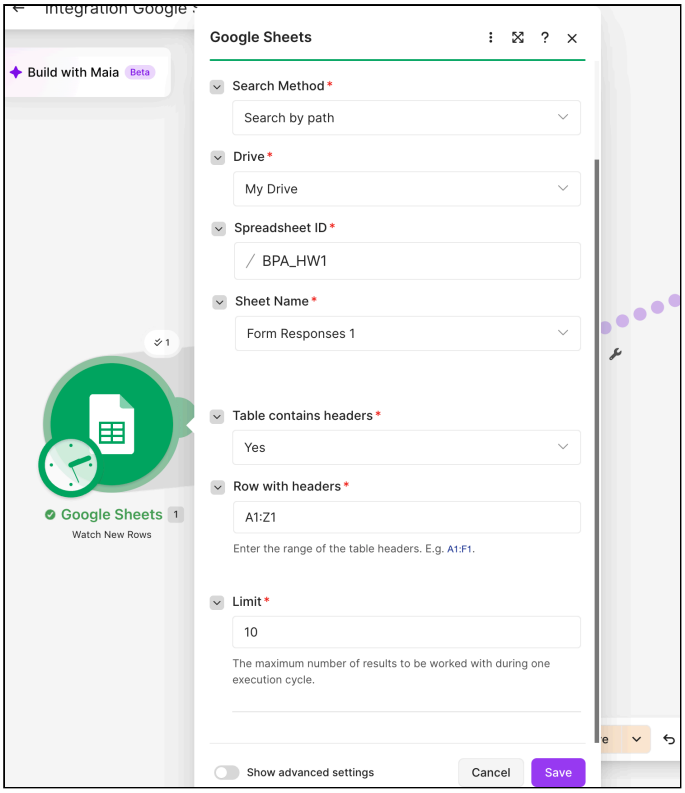
1. **Search Method:** Search by path · **Drive:** My Drive
2. **Spreadsheet ID:** press the folder icon and pick **your** form's sheet.
3. **Sheet Name: Form Responses 1**
4. **Table contains headers:** Yes · **Row with headers:** A1:Z1
5. **Limit: 10**
6. Save.

WHAT "LIMIT" MEANS

Limit = how many **new rows** Make takes in one run. Each row is one person's answers. With **10**, all your classmates' answers come in together, and the aggregator can join them into one report.

MAKE REMEMBERS THE LAST ROW

After a run, Make remembers which row it used last. The next run only looks **after** that row. That is why you choose **All** before each Run once (page 7).



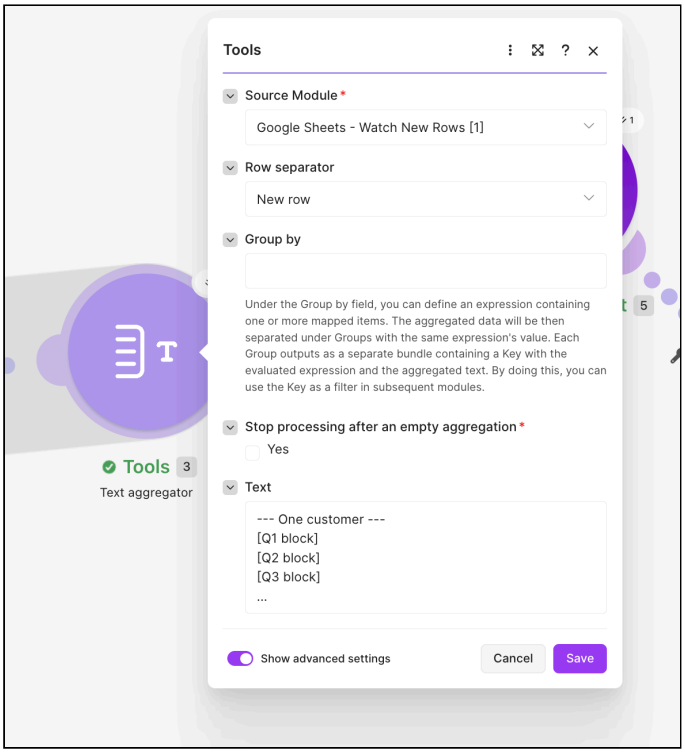
The Sheets point. Your sheet name goes where BPA_HW1 is.

4 Text aggregator — what the AI will read

1. **Source Module:** Google Sheets – Watch New Rows [1]
2. **Row separator:** New row
3. **Stop processing after an empty aggregation:** tick **Yes** — no answers, no empty email.
4. **Text:** delete the placeholder lines. Keep the first line `--- One customer ---`. Under it, click your question blocks one by one from the list on the right (B, C, D …).

```
--- One customer ---
Price you would pay: [your column E block]
Who you are: [your column F block]
...
```

Write a short label before each block. The AI reads the label and knows what the answer is about.



In the template the Text box shows placeholders like **[Q1 block]**. Delete them — they are instructions, not data.

5 Make AI Toolkit — first count, then write

The template prompt already asks the AI to **count first** (“4 of 5”) and **then** write a three-sentence report. You change one line: **My business**.

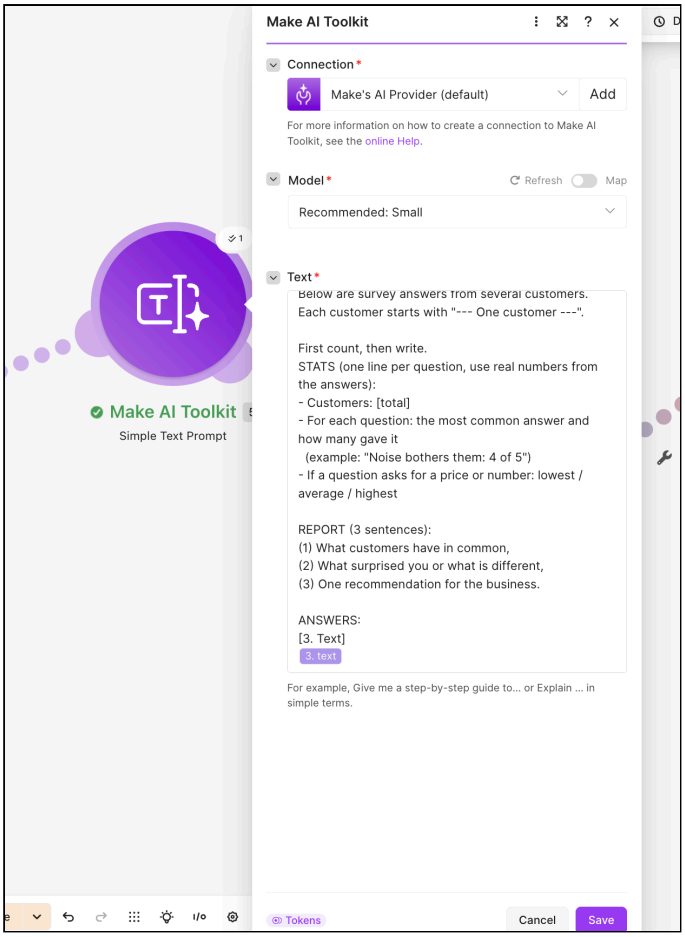
```
You are helping a small business owner.
My business: [ONE SENTENCE ABOUT YOUR BUSINESS]

Below are survey answers from several customers.
Each customer starts with "--- One customer ---".

First count, then write.
STATS (one line per question, real numbers):
- Customers: [total]
- For each question: the most common answer
  and how many gave it (e.g. "4 of 5")
- If a question asks for a price or number:
  lowest / average / highest

REPORT (3 sentences):
(1) What customers have in common,
(2) What surprised you or what is different,
(3) One recommendation for the business.

ANSWERS:
[the aggregator's Text block]
```



At the end there must be only the purple **3. text** block. In this screenshot the words “[3. Text]” are still there too — delete words like that.

REAL MISTAKE — WHAT HAPPENS IF YOU FORGET TO CHANGE THE BRACKETS

The professor’s first test email started with **“ONE SENTENCE ABOUT YOUR BUSINESS STATS: …”** and the infographic title said **“ONE SENTENCE ABOUT YOUR BUSINESS”**. The AI copies whatever you leave in the prompt. **Every [BRACKET] must be replaced or deleted.**

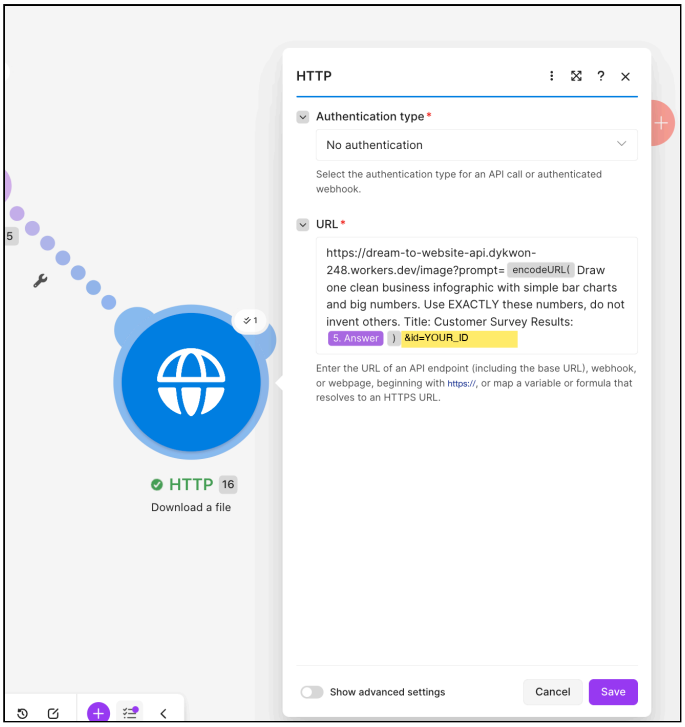
Make it better — change the three report questions for your business

Your business	Try this as question (2)
Café / food	Which time of day do most customers want to come, and why might that be?
Online shop	Which customers would pay more than the average, and what do they have in common?
Service / class	What is the biggest worry that stops customers from trying it?

Keep the report short (3 sentences). The picture prompt can hold only about **600 characters** — a long report gets cut.

6 HTTP · Download a file — the picture

1. **Authentication type:** No authentication
2. **URL:** keep everything. Change only the end: `&id=YOUR_ID` → `&id=`**your student ID** (the ID you log in with on the class site).
3. Inside `encodeURL( … )` is your **picture prompt** + the AI's **5. Answer** block. You may rewrite the prompt words — keep the block.



Your student ID goes where YOUR_ID is. Your ID can make 10 images a day — test with 1–2 answers, not 20.

WHY encodeURL

A web address cannot hold spaces, line breaks or "&". `encodeURL` turns the AI's text into a safe address. Without it, the address breaks and you get **Bad Request**.

WHY "DRAW …" MUST COME FIRST

If the prompt is only a report, the image model often answers with text and no picture — error **Bad Gateway / image_failed**. Always start with an instruction to draw.

Make the picture fun — the five-part recipe

```
Draw a [STYLE] showing [WHO] in [WHERE].
Mood: [MOOD]. Show these numbers clearly, exactly as written:
[the AI's Answer block]
```

STYLE

infographic with bar charts · 3-panel comic strip · retro travel poster · newspaper front page · TV weather report · video-game level map · LEGO scene · recipe card

WHO & WHERE

Use your Part 2 answers: "a few friendly customers aged 40+ at home". Tell it **several** customers — the picture is about the group, not one person.

MOOD

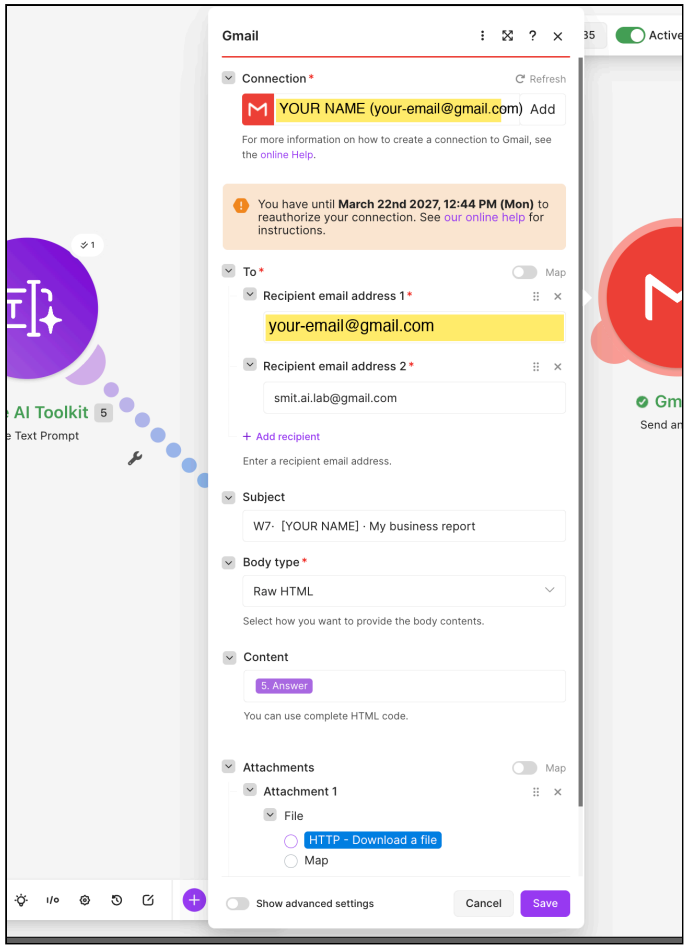
calm · energetic · funny · serious · cosy. Add one surprise: "a cat is checking the price tag". Small jokes make people read the numbers.

Three rules that make it better

1. **Numbers first, decoration second.** Write "Use EXACTLY these numbers, do not invent others." The image model sometimes changes numbers — check them.
2. **One idea per picture.** If it gets messy, ask for fewer panels: "three panels only".
3. **Change one word at a time.** Change only the STYLE, run again, compare. Keep the version people understand fastest.

7 Gmail — to you and to the class inbox

1. **Connection:** your Google account (page 3).
2. **Recipient 1: your** email address.
3. **Recipient 2:** keep **smit.ai.lab@gmail.com** — this is how you submit.
4. **Subject:** `W7 · your name · My business report` (Week 8 midterm: start with **W8**).
5. **Content:** the AI's **5. Answer** block.
6. **Attachments** → **File: click the circle** next to **HTTP – Download a file** so it is filled. A blue name with an empty circle means **not chosen**.
7. Save.

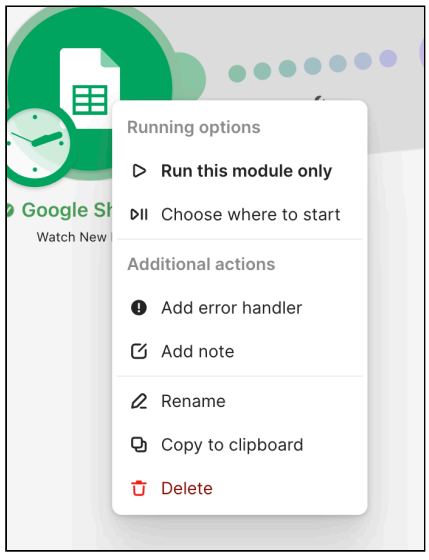


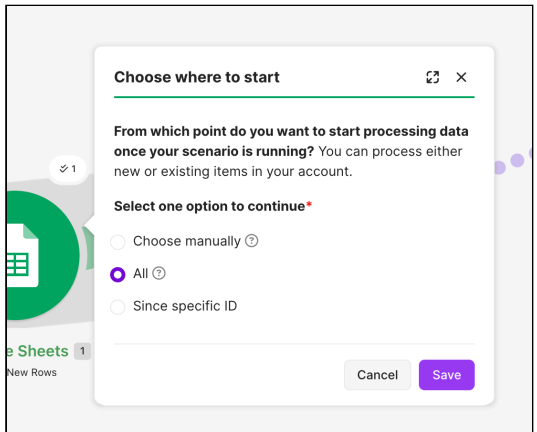
Look at the bottom: the circle next to "HTTP – Download a file" is still empty here. Click it.

8 Test → clean → send → Run once

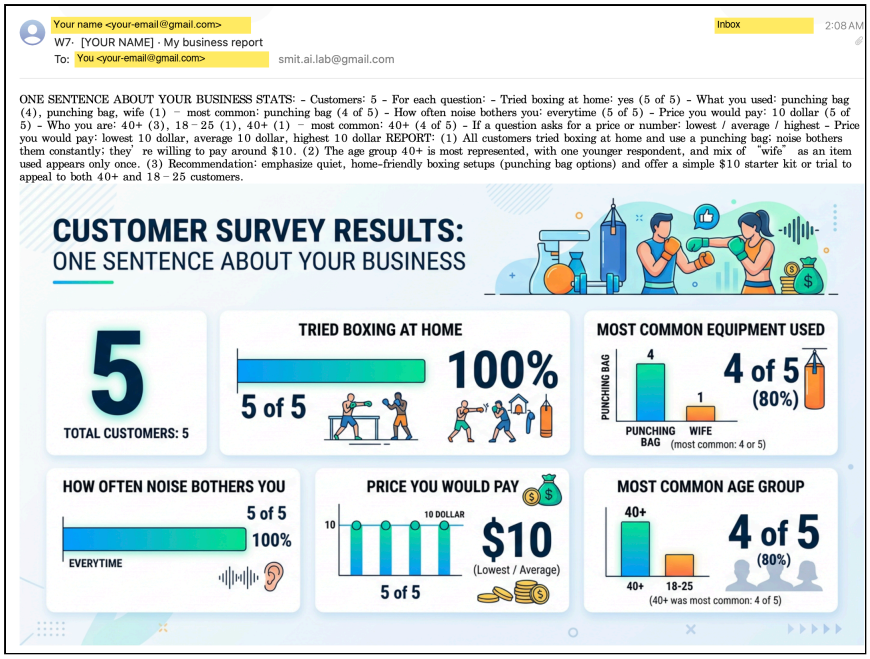
1. **Test:** answer your own form **twice**.
2. Right-click the **Sheets** point → **Choose where to start** → **All** → Save.
3. Press **Run once**. Wait. Check your email: one report + one picture.
4. **Clean:** delete your 2 test rows in the sheet (keep row 1, the questions). In the form: Responses → ⋮ → Delete all responses.
5. **Send** the responder link to **5 classmates**. Answer 5 of theirs too.
6. When 5 answers are in: **Choose where to start** → **All** → **Run once**.

Choose **All** every time before Run once. Make remembers the last row it used, so without All a second run finds "nothing new" and sends nothing.





What a good result looks like



One email. On top: STATS with real counts (“punching bag 4 of 5”, “40+ 4 of 5”, “$10 lowest / average / highest”) and a 3-sentence report. Below: the infographic drew the same numbers. Mistake to notice: “ONE SENTENCE ABOUT YOUR BUSINESS” was never replaced.

Check the picture against your sheet: count one question by hand. Different number? The image model invented it — run again.

When it does not work

You see	Do this
No email at all	Did you press **Save**? Did you choose **All** before Run once? Is there at least one row in Form Responses 1?
Email but no picture attached	Gmail → Attachments → click the **circle** next to HTTP – Download a file.
HTTP: **Bad Request**	Put the Answer block **inside** encodeURL(…). Check that `&id=` has your ID.
HTTP: **Bad Gateway** · image_failed	Start the picture prompt with “Draw …”. Make the AI report shorter.
HTTP: “Student ID not found” (403)	Use the ID you log in with on the class site.
HTTP: “Image limit reached” (429)	Your daily images are used up. Try again tomorrow (the count resets at 09:00 Seoul time).
Sheets / Gmail: connection or permission error	Connection → Add → your account → **tick every box** → Allow.
Report talks about only one person	Limit must be 10, and the Text aggregator must sit between Sheets and the AI.

ONE RUN, ONE EMAIL — WHY?

Each row is one bundle. Without the aggregator, every point after Sheets runs once **per row** — 5 answers, 5 emails. The **Text aggregator joins the bundles into one**, so the AI sees everyone at once and can count. That is how “4 of 5” becomes possible.

Week 8 midterm — PEOPLE (no class, Oct 21): the exam is the value you create with real people. Tell people about your survey, ask them honestly — at least 5 classmates answer your survey → one report email with the infographic arrives at smit.ai.lab@gmail.com (subject starts with **W8**) → answer the questions in the AI assistant chat on the class site. Deadline **Oct 21, 18:00**. Change the template to fit your business and you get an extra ★. **Today is practice** with your own test answers. **During the week** you collect 5 classmates — and if you did not finish building today, you finish it now. On Oct 21 the assistant is open on **every week’s page**: submit Week 7 and **every week you missed**.